

Advanced Programming Practices

MTech 2026

Lab - 10

General Instructions:

- Write one .c file per problem in the assignment, renamed to RollNo_APP10_Qn.c (e.g. ABC2026001_APP10_Q1.c).
 - Every program must compile without warnings using the gcc compiler.
 - Free any memory you allocate. Programs that leak memory or access out-of-bounds indices will lose marks even if the visible output looks correct.
 - Test each program with at least one normal case, one edge case (e.g. size 0 or 1, all-equal elements), and one larger randomly generated case.
 - Submit all the .c files on Google Classroom before the due time, any late submissions will be penalized or discarded.
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1. Write a C program to sort an array containing both positive and negative integers such that negative numbers appear first, followed by positive numbers. Maintain ascending order within each group.
2. Write a C program to sort an array using Bubble Sort and then remove all duplicate elements from the sorted array. Display the resulting array.
3. Write a C program to sort the array of integers using quicksort or mergesort algorithm.
4. Write a C program to sort an array using Selection Sort and display the array after every pass of the sorting process.
5. Write a C program to sort an array of integers according to their absolute values. For example, -2, 1, -5, 3 should be arranged as 1, -2, 3, -5.